=== UNIT 1: DC CIRCUIT ANALYSIS & NETWORK THEOREMS ===

>> 1.1 Fundamental Circuit Quantities and Basic Laws

Direct Current (DC) circuits form the foundational basis for power supplies, instrumentation, and bioelectrical measurement interfaces.
* Circuit Definitions: Node (junction where 2 or more elements connect), Essential Node (junction of 3 or more elements), Branch (path connecting two nodes containing an element), Mesh (a loop containing no other loops inside it).
* Ohm's Law: Current I through a linear conductor between two points is directly proportional to voltage V: V = I * R, where R is resistance in Ohms (Omega). Resistivity relation: R = rho * L / A.
* Kirchhoff's Current Law (KCL): The algebraic sum of currents entering any electrical node is identically zero: sum I_entering = sum I_leaving. Grounded in the Law of Conservation of Charge.
* Kirchhoff's Voltage Law (KVL): The algebraic sum of all potential differences around any closed circuit loop is identically zero: sum V_drops = sum V_rises. Grounded in the Law of Conservation of Energy.
* Mesh (Loop) Analysis: Based on KVL. Formulates a system of linear equations [R_mesh] * [I_mesh] = [V_mesh] where R_ii is the self-resistance of mesh i, and R_ij is the mutual resistance shared between mesh i and mesh j.
* Nodal Analysis: Based on KCL. Formulates [G_node] * [V_node] = [I_node] where diagonal elements are the sum of conductances attached to node i, and off-diagonal elements are negative mutual conductances.

>> 1.2 Network Theorems for Linear Bilateral Circuits

* Superposition Theorem: In any linear network containing multiple independent sources, the voltage across (or current through) any element is the algebraic sum of the voltages (or currents) caused by each independent source acting alone, with all other independent sources deactivated (voltage sources replaced by short circuits, current sources replaced by open circuits). Dependent sources remain active.
* Thevenin's Theorem: Any linear, two-terminal bilateral network consisting of sources and resistances can be completely replaced by an equivalent circuit containing a single ideal voltage source (V_th) in series with an equivalent resistance (R_th). V_th is the open-circuit voltage across terminals (V_oc); R_th is the equivalent resistance seen looking into the terminals with all independent sources deactivated.
* Norton's Theorem: Dual of Thevenin's theorem. Any two-terminal linear network can be replaced by a single ideal current source (I_N) in parallel with an equivalent resistance (R_N). I_N is the short-circuit current across terminals (I_sc); R_N = R_th. Source transformation relation: V_th = I_N * R_th.
* Maximum Power Transfer Theorem: A linear DC network delivers maximum electrical

power to a variable load resistor R_L when the load resistance equals the Thevenin equivalent resistance of the network: R_L = R_th. At maximum power transfer, the efficiency is exactly 50%: P_max = (V_th)^2 / (4 * R_th).

[SOLVED PROBLEM 1: Mesh Current Analysis of Two-Mesh DC Circuit]

Problem: In a two-mesh circuit, Mesh 1 has a 20V battery in series with a 4 Omega resistor, connected to a common branch containing a 6 Omega resistor. Mesh 2 connects the 6 Omega common resistor to an 8 Omega resistor and a 12V battery. Determine the loop currents I1, I2 and the current through the 6 Omega shared branch.

Solution Step 1: Set up KVL mesh equations:

Mesh 1 (clockwise current I1):

20 - 4*I1 - 6*(I1 - I2) = 0 => 10*I1 - 6*I2 = 20.

Mesh 2 (clockwise current I2):

- 6*(I2 - I1) - 8*I2 - 12 = 0 => - 6*I1 + 14*I2 = -12.

Solution Step 2: Solve the system using Cramer's rule:

Delta = det([10 -6 ; -6 14]) = (10)(14) - (-6)(-6) = 140 - 36 = 104.

Delta_1 = det([20 -6 ; -12 14]) = (20)(14) - (-12)(-6) = 280 - 72 = 208.

Delta_2 = det([10 20 ; -6 -12]) = (10)(-12) - (-6)(20) = -120 + 120 = 0.

Solution Step 3: Compute currents:

I1 = Delta_1 / Delta = 208 / 104 = 2.0 Amperes.

I2 = Delta_2 / Delta = 0 / 104 = 0.0 Amperes.

Solution Step 4: Branch current through 6 Omega resistor:

I_6Omega = I1 - I2 = 2.0 - 0.0 = 2.0 Amperes (flowing downwards).

Check: Voltage drop across 6 Omega = 2.0 A * 6 Omega = 12 V. In Mesh 2, this exactly balances the 12V battery, confirming I2 = 0.

[SOLVED PROBLEM 2: Thevenin Equivalent and Maximum Power Transfer]

Problem: For a bridge circuit where terminals A-B are connected across a variable load R_L: A 24V source feeds a parallel divider of (R1 = 6 Omega, R2 = 3 Omega) on the left and (R3 = 4 Omega, R4 = 8 Omega) on the right. Terminals A and B are between the dividers. Find: (1) Thevenin voltage V_th, (2) Thevenin resistance R_th, (3) Value of R_L for maximum power, (4) P_max.

Solution Step 1: Calculate Open-Circuit Voltage V_th (V_A - V_B):

V_A = 24 * [R2 / (R1 + R2)] = 24 * [3 / (6 + 3)] = 24 * (3 / 9) = 8.0 V.

V_B = 24 * [R4 / (R3 + R4)] = 24 * [8 / (4 + 8)] = 24 * (8 / 12) = 16.0 V.

V_th = | V_A - V_B | = | 8.0 - 16.0 | = 8.0 Volts (Terminal B is positive relative to A).

Solution Step 2: Calculate Thevenin Resistance R_th (deactivate 24V voltage source to short circuit):

Seen from terminals A-B, R1 is in parallel with R2, and R3 is in parallel with R4:

R_A = (R1 * R2) / (R1 + R2) = (6 * 3) / (6 + 3) = 18 / 9 = 2.0 Omega.
R_B = (R3 * R4) / (R3 + R4) = (4 * 8) / (4 + 8) = 32 / 12 = 2.67 Omega.
R_th = R_A + R_B = 2.0 + 2.67 = 4.67 Omega (or 14/3 Omega).
Solution Step 3: Optimal Load for Maximum Power:
By Maximum Power Transfer Theorem: R_L = R_th = 4.67 Omega.
Solution Step 4: Maximum Power Delivered P_max:
P_max = (V_th)^2 / (4 * R_th) = (8.0)^2 / (4 * (14/3)) = 64 / (56 / 3) = 192 / 56 = 3.43 Watts.

[SOLVED PROBLEM 3: Application of Superposition Theorem]

Problem: In a linear circuit, a 12V voltage source and a 2A current source simultaneously power a resistor network: The 12V source connects to R1 = 4 Omega in series with R2 = 4 Omega. The 2A current source is connected in parallel across R2. Find the total current through R2 using Superposition.
Case 1: 12V Voltage Source acting ALONE (Current source open-circuited):
Circuit is a simple series loop with R1 and R2:
R_total = R1 + R2 = 4 + 4 = 8 Omega.
I2' = V / R_total = 12 / 8 = 1.5 Amperes (flowing downwards through R2).
Case 2: 2A Current Source acting ALONE (Voltage source short-circuited):
The 12V source is replaced by a wire, placing R1 (4 Omega) in parallel with R2 (4 Omega):
Current from the 2A source splits equally between two equal 4 Omega branches:
I2'' = 2 * [R1 / (R1 + R2)] = 2 * [4 / (4 + 4)] = 2 * (1/2) = 1.0 Ampere (flowing downwards).
Total Current by Superposition:
I_total = I2' + I2'' = 1.5 A + 1.0 A = 2.5 Amperes.
Power in R2: P = (I_total)^2 * R2 = (2.5)^2 * 4 = 6.25 * 4 = 25.0 Watts.

[SOLVED PROBLEM 4: Norton Equivalent Circuit and Current Source Transformation]

Problem: Derive the Norton Equivalent Circuit for a DC source of 30V having an internal resistance of 5 Omega connected to a parallel branch of 10 Omega. Determine the current delivered to a 15 Omega load resistor connected across the output terminals.
Solution Step 1: Calculate Norton Short-Circuit Current I_N:
Short-circuit the load terminals. The 10 Omega resistor is bypassed by the short circuit.
The short-circuit current is limited only by the 5 Omega resistor:
I_sc = I_N = V / R_internal = 30 V / 5 Omega = 6.0 Amperes.
Solution Step 2: Calculate Norton Resistance R_N:
Deactivate 30V source to short circuit. Looking into output terminals:
R_N = 5 Omega in parallel with 10 Omega = (5 * 10) / (5 + 10) = 50 / 15 = 3.33 Omega

(10/3 Omega).
Solution Step 3: Connect Load R_L = 15 Omega to Norton Equivalent:
Current divider rule determines load current I_L:
I_L = I_N * [R_N / (R_N + R_L)] = 6.0 * [(10/3) / ((10/3) + 15)] = 6.0 * [(10/3) / (55/3)]
I_L = 6.0 * (10 / 55) = 60 / 55 = 1.09 Amperes.
Voltage Across Load: V_L = I_L * R_L = 1.091 A * 15 Omega = 16.36 Volts.

>> 1.3 University Review & Examination Questions

* Q1: State and prove the Maximum Power Transfer Theorem for DC networks.
Answer: Statement: A resistive network delivers maximum power to a variable load resistance R_L when R_L is equal to the Thevenin equivalent resistance R_th of the network. Proof: Replace the network by its Thevenin equivalent (V_th in series with R_th). The load current is I_L = V_th / (R_th + R_L). The power delivered to the load is P_L = (I_L)^2 * R_L = [(V_th)^2 * R_L] / (R_th + R_L)^2. To find the maximum power, differentiate P_L with respect to R_L and set dP_L / dR_L = 0: dP_L / dR_L = (V_th)^2 * [(R_th + R_L)^2 * (1) - R_L * 2*(R_th + R_L)] / (R_th + R_L)^4 = 0. Simplifying numerator: (R_th + R_L) - 2*R_L = 0 => R_th - R_L = 0 => R_L = R_th. Hence proved.
* Q2: Explain the step-by-step procedure to determine Thevenin's equivalent resistance (R_th) when a circuit contains both independent and dependent sources.
Answer: When dependent sources are present, R_th cannot be found by simply deactivating sources. Procedure: (1) Determine the open-circuit voltage V_oc across terminals A-B using standard nodal or mesh analysis while keeping all sources active. (2) Connect an ideal short circuit across A-B and compute the short-circuit current I_sc. (3) Thevenin resistance is given by R_th = V_oc / I_sc. Alternatively: Deactivate only the INDEPENDENT sources, connect an external test source (either V_test = 1V or I_test = 1A) across terminals A-B, determine the resulting current I_test (or voltage V_test), and calculate R_th = V_test / I_test.

>> 1.4 Key Takeaways and Revision Points

* KCL represents charge conservation at nodes; KVL represents energy conservation around loops.
* Thevenin equivalent consists of V_th in series with R_th; Norton equivalent consists of I_N in parallel with R_N (where V_th = I_N * R_th).
* Maximum power transfer occurs at R_L = R_th with 50% efficiency (P_max = V_th^2 / 4*R_th).
* Superposition requires linear bilateral elements; deactivated voltage sources are shorted, current sources opened.